



This IR Remote Can Control Up To Eight Devices

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This project based on an 8-bit ATmega328P microcontroller (MCU) can control up to eight devices connected to its electromagnetic relays. The article also describes how to program an MCU to accept infrared (IR) remote control signals for controlling the relays.

Here, an ordinary TV remote control is used along with the 28-pin ATmega328P MCU in DIL package and Arduino bootloader to control the devices. Pin details of ATmega328P are given in Fig. 1.

Circuit and working

Block diagram of the project is shown in Fig. 2 and its complete circuit diagram is shown in Fig. 3. To understand the project, it is divided into following four sections:

1. Power supply unit
2. Infrared light receiver unit
3. Microcontroller unit and its programming
4. Relay driver output and power isolator unit

PARTS LIST

Semiconductors:

IC1	- 7812, 12V voltage regulator
IC2	- 7805, 5V voltage regulator
IC3	- ATmega328P MCU
IC4	- ULN2803 relay driver
LED1-LED9	- 5mm LED
D1-D4	- 1N4007 rectifier diode
IRRX1	- TSOP1738

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R9	- 1-kilo-ohm
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Capacitors:

C1	- 1000 μ F, 40V electrolytic
C2-C3	- 22pF ceramic disk

Miscellaneous:

X1	- 230V AC primary to 15V, 1A secondary transformer
XTAL1	- 16MHz crystal
CON1-CON8	- 2-pin connector
CON9	- 2-pin terminal connector
RL1-RL8	- 12V, single changeover relay
	- 28-pin IC base
	- 18-pin IC base
	- Heat-sink for IC1 and IC2

(All these are available from kitsnspares.com)

Power supply unit. This section includes a 230V AC primary to 15V, 1A secondary step-down transformer (X1). The secondary of this transformer is connected to a bridge rectifier circuit formed by four IN4007 rectifier diodes (D1 to D4) and filter capacitor C1. Regulators 7812 (IC1) and 7805 (IC2) are used to get required 12V and 5V supplies, respectively.

Infrared light receiver unit. An IR module TSOP1738 (IRRX1) is used to receive infrared light signal from any TV IR remote control. The IR remote control signals are sent to IC3 microcontroller's pin 18 for processing and controlling the relays as per program and calibration of the controller unit.

Microcontroller unit and programming. This unit is the heart of the device. It has two parts: hardware and software.

The hardware receives the infrared light's hex-coded signals generated

by TV remote control and sends them to the MCU through infrared receiver module TSOP1738 (IRRX1). Output pin 3 of IRRX1 is connected to MCU's pin 18. The MCU's output pins are programmed to turn on or turn off using IR remote input signal. MCU's output pins 4, 5, 6, 11, 12, 13, 14 and 15 are connected to pins 1 through 8 of ULN2803 (IC4) relay driver unit. The output pins 18 through 11 of IC4 are connected to RL1 through RL8 relays, respectively as shown in Fig. 3.

The MCU can be programmed by following the steps below:

1. Download Arduino IDE software and install it in the PC
2. Connect USB cable between Arduino board and the PC
3. Open Arduino IDE by clicking the Arduino icon from Desktop. (You will get Arduino programming/sketch window as shown in Fig. 4.)

After installing Arduino IDE in the PC, install IR remote libraries. For this use IRremote-2.0.1 and IRremote-2.2.3 folders included in this project. Copy these folders under Arduino/Libraries in a Windows PC. Alternatively, you can install the libraries from Arduino IDE as explained below:

1. Select Sketch $\rightarrow$ Include Library $\rightarrow$ Manage Libraries option
2. Type IR Remote and press Enter

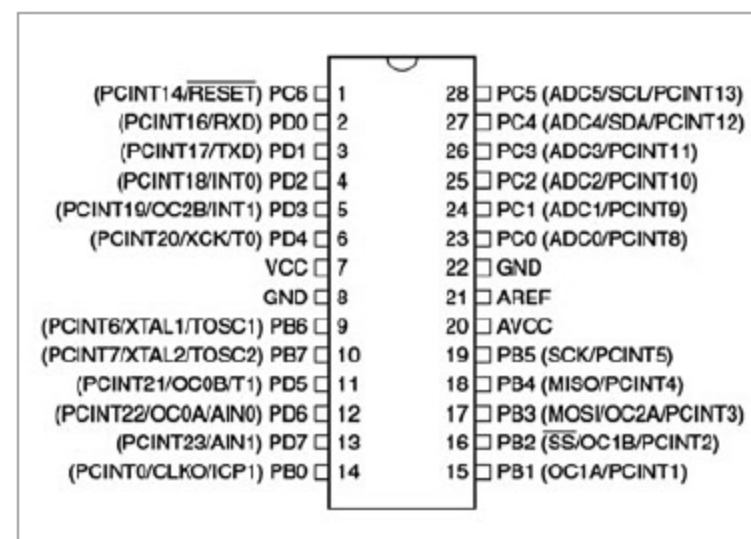


Fig. 1: ATmega328P pin details

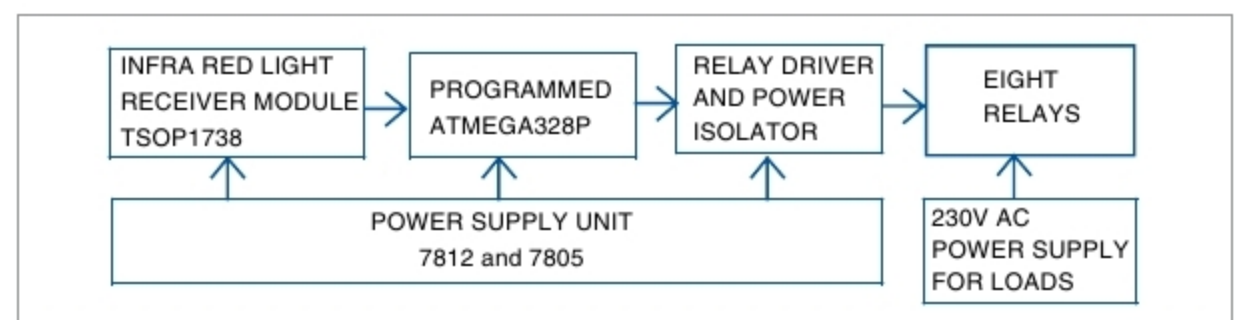


Fig. 2: Block diagram of the project